1. **What is the output of 5 & 3?**
   A) 7
   B) 1
   C) 0
   D) 5
   Answer: B

2. **What is the output of 5 | 2?**
   A) 7
   B) 0
   C) 1
   D) 3
   Answer: A

3. **What is the output of 5 ^ 5?**
   A) 0
   B) 10
   C) 1
   D) 5
   Answer: A

4. **What is the result of ~5?**
   A) 4
   B) -4
   C) -5
   D) -6
   Answer: D

5. **What is the result of 1 << 2?**
   A) 2
   B) 4
   C) 8
   D) 1
   Answer: B

6. **What is the result of 8 >> 2?**
   A) 2
   B) 4
   C) 8
   D) 1
   Answer: A

7. **Which operator gives bitwise XOR in Python?**
   A) &
   B) |
   C) ^
   D) ~
   Answer: C

8. **Which operator flips all bits of a number?**
   A) |
   B) &
   C) ~
   D) ^
   Answer: C
9. **What is the binary of 5 & 3 result?**
   A) 0b101
   B) 0b001
   C) 0b111
   D) 0b000
   Answer: B
10. **What is 0b1010 | 0b0101?**
   A) 0b0000
   B) 0b1001
   C) 0b1111
   D) 0b1100
   Answer: C

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11. **What is 12 & 10?**
   A) 10
   B) 8
   C) 12
   D) 2
   Answer: B
12. **What is 12 | 10?**
   A) 12
   B) 10
   C) 14
   D) 8
   Answer: C
13. **What is 12 ^ 10?**
   A) 6
   B) 4
   C) 2
   D) 10
   Answer: A
14. **What is ~(-5)?**
   A) -6
   B) 4
   C) 5

D) -4

Answer: B

15. **What is the output of 3 << 3?**

A) 6

B) 9

C) 24

D) 12

Answer: C

16. **What is the output of 32 >> 4?**

A) 4

B) 2

C) 1

D) 8

Answer: D

17. **If x = 7, what is x & (x-1)?**

A) 6

B) 7

C) 0

D) 1

Answer: A

18. **If x = 8, what is x & (x-1)?**

A) 8

B) 0

C) 7

D) 1

Answer: C

19. **What does x & -x give for positive x?**

A) Largest power of 2 less than x

B) Smallest power of 2 dividing x

C) x-1

D) Always 0

Answer: B

20. **What is bool(4 & 0)?**

A) True

B) False

C) 1

D) 4

Answer: B

Hard (21–30)

21. **What is ((5 ^ 3) & (2 | 1))?**
    A) 7
    B) 0
    C) 1
    D) 2
    Answer: C

22. **What is ((~3) & 7)?**
    A) 4
    B) 5
    C) 3
    D) 7
    Answer: A

23. **If x=11, what is (x >> 1) + (x & 1)?**
    A) 5
    B) 6
    C) 11
    D) 7
    Answer: B

24. **What is bin((5 << 1) | 1)?**
    A) 0b1011
    B) 0b101
    C) 0b1101
    D) 0b111
    Answer: A

25. **Which expression checks if n is a power of 2 (n>0)?**
    A) n & (n+1) == 0
    B) n & (n-1) == 0
    C) n | (n-1) == 0
    D) n ^ (n-1) == 0
    Answer: B

26. **What is (-1 >> 1) in Python?**
    A) 0
    B) 1
    C) -1
    D) -2
    Answer: C

27. **What is (1 << 31) >> 31 in Python (on 32-bit)?**
    A) 1
    B) 0
    C) -1
    D) Undefined
    Answer: C

28. **What is ((8 >> 1) ^ (2 << 2))?**
    A) 2
    B) 4
    C) 8
    D) 0
    Answer: D

29. **What is 3 & 5 | 6 ^ 4?**
    A) 5
    B) 7
    C) 3
    D) 6
    Answer: B

30. **What is ((15 & 7) << 2) | (3 ^ 1)?**
    A) 28
    B) 30
    C) 31
    D) 27
    Answer: C

DS

1. **What is the output of list("abc")?**
   A) "abc"
   B) ['a','b','c']
   C) ['abc']
   D) ['a b c']
   Answer: B

2. **What is len((1,2,(3,4)))?**
   A) 3
   B) 4
   C) 2
   D) 5
   Answer: A

3. **Which is mutable?**
   A) tuple
   B) string
   C) list
   D) frozenset
   Answer: C

4. **What is the output of set([1,2,2,3])?**
   A) {1,2,2,3}
   B) {1,2,3}

C) [1,2,3]
D) (1,2,3)
Answer: B

5. **What is {}.__class__?**
   A) <class 'list'>
   B) <class 'dict'>
   C) <class 'set'>
   D) <class 'tuple'>
   Answer: B

6. **What is type(set())?**
   A) list
   B) dict
   C) set
   D) tuple
   Answer: C

7. **What is len({'a':1, 'b':2})?**
   A) 2
   B) 1
   C) 4
   D) 3
   Answer: A

8. **What is [[]]*3?**
   A) 3 different empty lists
   B) 3 references to same list
   C) Error
   D) [[], [], None]
   Answer: B

9. **Which data structure does collections.deque implement?**
   A) Stack
   B) Queue/Double-ended queue
   C) Heap
   D) Set
   Answer: B

10. **Which structure ensures unique elements and is unordered?**
    A) dict
    B) list
    C) set
    D) tuple
    Answer: C

Medium (11–20)

11. **What is a = [1,2,3]; b=a; b.append(4); print(a)?**
    A) [1,2,3]
    B) [1,2,3,4]
    C) [1,2,3] [1,2,3,4]
    D) Error
    Answer: B

12. **What is tuple([1,2,3])?**
    A) (1,2,3)
    B) ['1','2','3']
    C) (1)
    D) Error
    Answer: A

13. **What is the output of {1,2,3} & {2,3,4}?**
    A) {1,2,3,4}
    B) {2,3}
    C) {1,4}
    D) True
    Answer: B

14. **What is sorted({'x':2,'a':1})?**
    A) [2,1]
    B) ['x','a']
    C) ['a','x']
    D) ['x':2,'a':1]
    Answer: C

15. **Which of these is not hashable?**
    A) int
    B) str
    C) tuple of tuples
    D) list
    Answer: D

16. **What is max({'a':5,'b':3})?**
    A) 5
    B) 'a'
    C) 'b'
    D) 3
    Answer: B

17. **What is dict.fromkeys('abc',0)?**
    A) {'a':0,'b':0,'c':0}
    B) {'abc':0}
    C) {'a','b','c':0}
    D) Error
    Answer: A

18. **What is [1,2]+[3,4]*2?**
    A) [1,2,3,4,3,4]
    B) [1,2,6,8]
    C) Error
    D) [1,2,3,4,2]
    Answer: A
19. **What is len(set("balloon"))?**
    A) 7
    B) 6
    C) 5
    D) 4
    Answer: D
20. **Which method removes and returns last element of list?**
    A) pop()
    B) remove()
    C) del()
    D) discard()
    Answer: A

Hard (21–30)

21. **What is a=[1,2,3]; b=a[:]; b.append(4); print(a)?**
    A) [1,2,3,4]
    B) [1,2,3]
    C) [1,2,3][1,2,3,4]
    D) Error
    Answer: B
22. **What is d={'x':1,'y':2}; d.update({'y':3,'z':4}); print(d)?**
    A) {'x':1,'y':2}
    B) {'x':1,'y':3,'z':4}
    C) {'x':1,'y':2,'z':4}
    D) {'x':1,'y':3}
    Answer: B
23. **What is d = {}; d[[]] = 1?**
    A) {[]:1}
    B) Error
    C) {'[]':1}
    D) {'':1}
    Answer: B
24. **What is d = {}; d[(1,2)] = 3?**
    A) {(1,2):3}
    B) {[1,2]:3}

C) Error
D) {'(1,2)':3}
Answer: A

25. **Which expression reverses a list L in-place?**
A) reversed(L)
B) L[::-1]
C) L.reverse()
D) list(reversed(L))
Answer: C

26. **What is s = {1,2,3}; s.add((4,5)); print(len(s))?**
A) 3
B) 4
C) 2
D) Error
Answer: B

27. **What is L=[[]]*3; L[0].append(1); print(L)?**
A) [[1],[],[]]
B) [[1],[1],[1]]
C) [[1,1,1]]
D) Error
Answer: B

28. **What is [i for i in range(3)]*2?**
A) [0,1,2,0,1,2]
B) [0,1,2,2,1,0]
C) [0,1,2]*2
D) Error
Answer: A

29. **Which gives a shallow copy of a list L?**
A) copy.copy(L)
B) L.copy()
C) L[:]
D) All of these
Answer: D

30. **What is min({True:5, False:3})?**
A) 5
B) 3
C) False
D) True
Answer: C